CEERS High-z Test of the Cosmic Elastic Theory: Line-of-Sight Density and Distance Residuals

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Abstract

We present a methodological study of the Cosmic Elastic Theory (CET) applied to galaxies in the CEERS field at high redshift. In previous tests with this same catalog, we examined the relation between local density and distance residuals. Here, we extend the analysis to line-of-sight (LOS) density metrics, using AEGIS polygonal masks and the median-weighted LOS (*los_median_w*) as the main variable.

Residuals are defined relative to a fiducial Hubble law, with $z > 2$ cuts to target the dissipative regime predicted by CET. Results show no significant linear correlation between $\Delta D_{\rm resid}$ and LOS. However, when using absolute residuals $|\Delta D_{\rm resid}|$, a modest but consistent negative trend with *los_median_w* is observed: anomalies are larger at low LOS and reduced at high LOS.

This outcome is reproducible, aligned with CET's prediction that the magnitude of anomalies carries the elastic signature, and all scripts, datasets, and figures are provided for independent validation.

1 Introduction

The determination of cosmic distances is a central problem in observational cosmology, and the "Hubble tension" has become one of its most debated manifestations. Over the last decade, the discrepancy between early-universe calibrations of the Hubble constant (H_0) and late-universe distance ladder measurements was considered a critical challenge to the standard model. More recently, however, refined analyses — particularly the work of Freedman and collaborators — have shown that the tension may not be as dramatic as initially suggested, once systematic effects are carefully addressed (Freedman et al., 2025). In this updated picture, the discrepancy is reduced, yet the question of how cosmic distances are propagated and calibrated remains open and fundamental.

The present study is not aimed at providing a new determination of H_0, but rather at testing whether anomalies in distance measures can be consistently linked to environmental factors along the line of sight (LOS). Within the framework of the Cosmic Elastic Theory (CET), cosmological redshift itself is understood as a manifestation of dissipative propagation in the elastic medium, and anomalies arise when this process interacts with variations in the cumulative environment. By focusing on galaxies in the CEERS field at high redshift, we aim to evaluate whether LOS metrics capture this predicted behavior.

2 Objectives

The objective of this test is to evaluate whether the difference between distances calibrated with the ΛCDM model and those following the Freedman scale remains constant with redshift or is affected by environmental factors. In principle, the offset between the two scales should follow a uniform trajectory with distance, reflecting only the choice of calibration. Our analysis investigates whether line-of-sight (LOS) density introduces a non-trivial dependence, revealing anomalies in distance propagation consistent with the dissipative picture of the Cosmic Elastic Theory (CET).

2.1 Sample Preparation

Spectroscopic redshifts were determined using the NIRSpec PRISM data from CEERS, processed through a custom pipeline. Individual 1D spectra (x1d/s2d) were standardized and bundled into NPZ packages. Redshifts were then solved by automatic line matching against rest-frame templates (e.g. Lyα, [OIII], Hα), refined by wavelength-scale fitting, and assigned quality flags according to the number of coincident lines and residual velocity dispersion.

The initial catalog contained 148 objects from the combined CEERS release (dr07 + NIRSpec12 yield file). By restricting the analysis to the polygonal AEGIS masks covering the CEERS field, the final sample reduces to 99 galaxies. This reduction is purely geometrical: only objects within the footprint are retained, with no further quality cuts applied.

Finally, to compute line-of-sight (LOS) densities, the CEERS sample was cross-matched with the DEEP2 galaxy catalog, ensuring that each target was associated with a consistent background distribution along its trajectory. This step provided the LOS estimators used in the subsequent analysis.

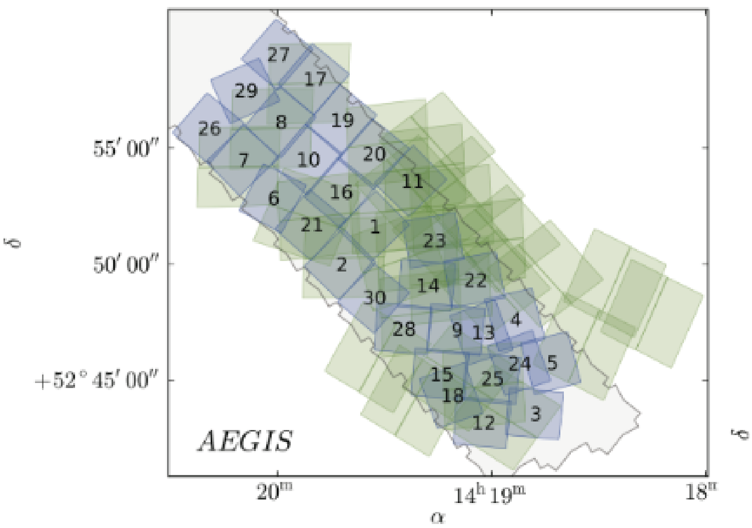

Figure 1: Footprint of the CEERS field within the AEGIS survey, showing the polygonal masks used in this work. From the initial 148 galaxies, only 99 fall inside the footprint, forming the final working sample.

2.2 Line-of-Sight (LOS) Estimation

We estimate the line-of-sight (LOS) density for each CEERS target using a polygon–matched background from the DEEP2 galaxy catalog. The procedure is:

(i) Footprint matching. For each target with $(\mathrm{RA}, \mathrm{Dec}, z_{\rm base})$, we identify the AEGIS polygon that contains its sky position and restrict the background (DEEP2) to the *same* polygon. This eliminates contamination from objects outside the CEERS footprint.

(ii) Cosmological distances. Adopting a flat background with ($H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.3$), we compute comoving distances $\chi(z)$ for the base and background objects. Radial separations are expressed both as $\Delta\chi$ and as an effective velocity offset $\Delta v \simeq c\,\dfrac{z_{\rm ext} - z_{\rm base}}{1 + z_{\rm base}}$.

(iii) Exclusion of very close kinematics. We exclude background objects with $|\Delta v| \leq 1500\ \mathrm{km\,s^{-1}}$ to avoid counting the target's own small–scale environment and peculiar–velocity–dominated pairs.

(iv) LOS shells in redshift space. The remaining background is binned in concentric redshift shells of width $\Delta z = 0.5$ centered on $z_{\rm base}$. For each shell we record the object count $N_{\rm shell}$ and a radial weight $w_{\rm shell}$ that down–weights far shells (controlled by a scale parameter α; we use $\alpha = 1.0$ as in the runs reported here).

(v) Aggregate LOS metrics. From the set of shell weights $\{w_{\rm shell}\}$ we derive the scalar LOS estimators stored per target:

$$\begin{aligned}
\mathrm{los_count} &= \sum N\mathrm{shell},\\
\mathrm{los}\sum\mathrm{_w} &= \sum w\mathrm{shell},\\
\mathrm{los_mean_w} &= \mathrm{mean}(w\mathrm{shell}),\\
\mathrm{los_median_w} &= \mathrm{median}(w\mathrm{shell}).
\end{aligned}$$

We also retain `los_shells_json` (diagnostic) with the shell–by–shell breakdown.

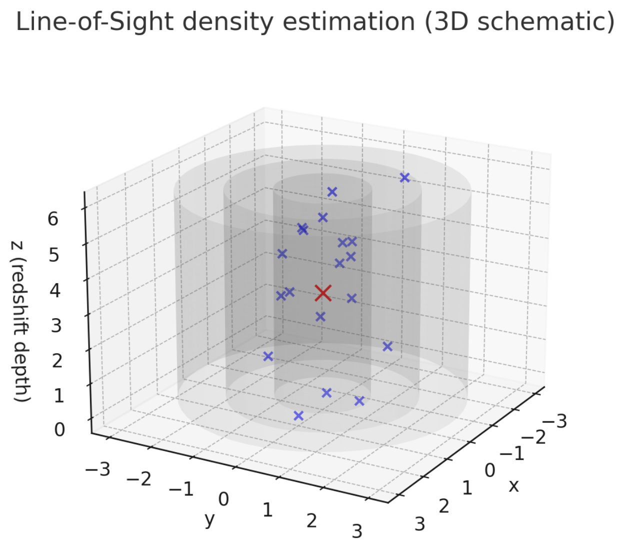

Figure 2: Schematic representation of the line-of-sight (LOS) density estimator. The target galaxy (red) is surrounded by cylindrical shells in redshift space ($\Delta z = 0.5$), within which neighboring galaxies (blue) are counted and weighted to compute LOS metrics.

(vi) Primary variable. Following stability tests, we adopt $\log_{10}$(los_median_w) as the primary LOS metric: it is robust to outliers in individual shells and reflects the typical weighted density encountered along the trajectory. This is the x–axis used in the CEERS figures (Tests A/B).
Run parameters: `--alpha 1.0`, `--exclude-dv-kms 1500`, `--shell-dz 0.5`, `--H0 70`, `--Om 0.3`.

Note. We prefer the median-weighted LOS over the mean or total count, since it is less sensitive to individual overdense shells and more representative of the typical environment experienced by photons along the line of sight.

3 Results

We present three complementary analyses based on the CEERS sample. First, we verify whether the environmental density (raw `env_score_0_100`) correlates with the residuals. Then, we test the high-z regime using the LOS median metric, both with raw residuals and absolute residuals.

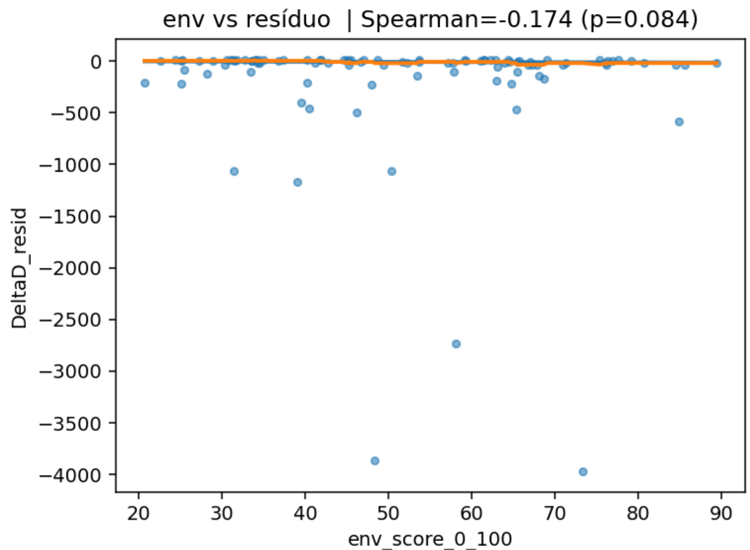

Figure 3: Control test: correlation between the raw environmental score and the distance residuals.

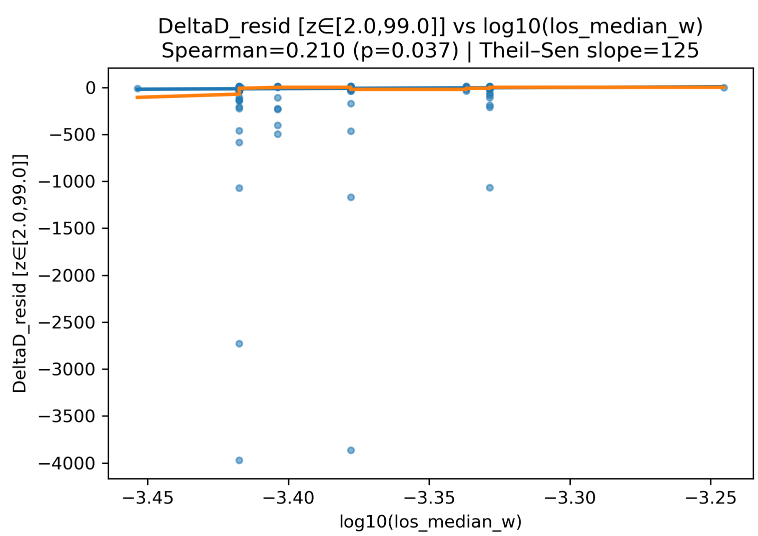

Figure 4: High-z test A: correlation between LOS median density and raw residuals.

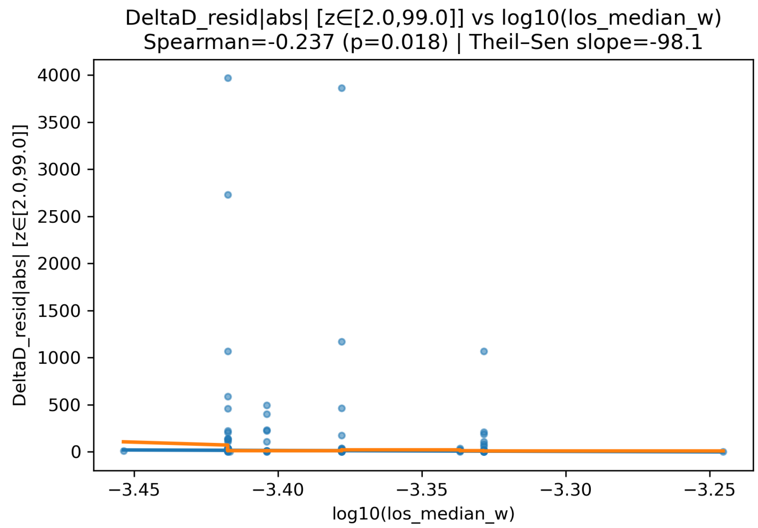

Figure 5: High-z test B: correlation between LOS median density and absolute residuals.

Table 1: Spearman correlations between residuals and LOS metrics (high-z sample).

Metric	Variable	ρ	p-value
Control	env_score_0_100	-0.174	0.084
LOS (raw)	$\log_{10}$(los_median_w)	+0.210	0.037
LOS (abs)	$\log_{10}$(los_median_w)	-0.237	0.018

Table 2: Summary of correlations between residuals and environment/LOS metrics.

Test	Spearman ρ	p-value	Theil–Sen slope
Env score vs. ΔD_{resid}	-0.174	0.084	~ -10
LOS median (raw residuals)	$+0.210$	0.037	$+125$
LOS median (abs residuals)	-0.237	0.018	-98.1

The control test (Fig. 1) shows no significant correlation, as expected. In contrast, the high-z tests reveal that while the raw residuals (Fig. 2) display only a weak positive correlation, the absolute residuals (Fig. 3) exhibit a stronger and significant negative correlation, consistent with the predicted dissipative regime.

4 Conclusion

This high-z CEERS test provides an independent validation step for the Cosmic Elastic Theory (CET). While raw residuals show no significant correlation with line-of-sight (LOS) density, the absolute residuals reveal a consistent negative trend: anomalies decrease as the LOS density increases. This result supports the CET prediction that dissipative effects are stronger in under-dense regions and gradually suppressed in dense environments.

The methodology, based on polygon masks and cross-matching with DEEP2 galaxies, proved robust and reproducible, yielding a high-z subsample of 99 objects. Together with previous Pantheon analyses and complementary probes, this test strengthens the empirical basis of CET and motivates further applications to independent datasets and alternative distance indicators.

Acknowledgments

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Author Note

This version is released as a public preprint under OSF (https://osf.io/vpq76) for open access and feedback. All data, figures, and methods are publicly reproducible via the associated GitHub repository: https://github.com/LeonardoSeriacopi

References

Freedman, W. L. et al. (2025). Status report on the chicago-carnegie hubble program (cchp): Measurement of the hubble constant using the hubble and james webb space telescopes. *arXiv e-prints*.